



Increasing interprofessional education experiences and collaboration: Outcomes of a virtually-based continuing education program

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ABSTRACT

Purpose: An academic medical center implemented a yearlong virtual continuing education program to prepare health professions educators to develop, deploy and evaluate new interprofessional educational experiences for providers and learners.

Materials and methods: Primary program outcome measures were the number of participants who (1) completed the program, (2) presented a medical education topic to program participants, and (3) designed an interprofessional educational experience in collaboration with fellow participants or colleagues. Participation outcomes were measured by session attendance and completion of presentations on the medical education topic and designed interprofessional educational experience. Surveys administered at 3 time points assessed participants' perceptions of the program. Changes in knowledge, skills, and attitudes were analyzed using paired t-tests.

Results: Nineteen participants (100 %) from 7 professions achieved the primary program participation outcomes and met their own goals. There were statistically significant increases in knowledge and skills at the end of the program. High rates of implemented interprofessional educational experiences (57 %) were reported one year after program completion. Participants also identified program strengths and areas for improvement.

Conclusions: Longitudinal continuing education programs in virtual formats can support both individuals and teams of educators in their own professional development and in designing interprofessional education experiences.

1. Introduction

The advancement of interprofessional education curricula for health professions students and providers is dependent upon effective training delivered by knowledgeable and skilled educators.^{1,2} The continuing education needs of health professions educators to implement such learning experiences has been articulated in health professions literature.^{3,4} A 2017 review article summarized several in-person continuing education programs and their outcomes to exemplify the successes and lessons learned.⁵ A 2018 study of in-person workshops conducted in several countries concluded participants had improvements in knowledge and skills in the design of interprofessional learning experiences for health professions students.⁶ Recent reviews have also focused on evolving opportunities and challenges of professional development for contemporary issues, including online learning and the COVID-19 pandemic.^{7,8,9} The literature on virtual programming has primarily focused developing facilitators of online interprofessional education.^{10–12} Few studies have described virtual-based continuing education to prepare educators to develop interprofessional curricula.

Our institution had identified a strategic priority of advancing interprofessional education for its learners and providers to enhance

patient care. To support achievement of this strategic priority, the institution's academy for health professions educators and center for interprofessional education and care collaborated to establish a new continuing education program for health professions educators. Administrative leadership recognized the established collaborative practice skills among its educators as well as lack of opportunity for training on educational program planning specific to interprofessional education experiences (IEEs). The original program format was in-person then was transitioned to a virtual format following the onset of the COVID-19 pandemic. This change was made to ensure educators had an opportunity for continuing education to advance interprofessional education at a time when established in-person activities were less accessible.

The goal of the yearlong virtual continuing education program was to prepare health professions educators to develop, deploy and evaluate IEEs in the context of learning about, from, and with educators from other professions. The program would support both individual professional development and participants' design of an IEEs for learners and/or providers. The design of this program was informed by published key lessons for program success: longitudinal learning, relevant content, effective instructional methods, communities of practice, competency-

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based outcomes, and education linked with practice within home institutions.^{2,3,13} Given the online format, best practices for virtual continuing education were also applied.^{10–12} The program was implemented from January–December 2021 within the Duke University School of Medicine.

Program evaluation aimed to assess the degree to which the objectives were achieved. This evaluation was particularly important given the unique nature of this program at the institution and the scarcity of literature describing longitudinal, virtual, interprofessional continuing education programs. The purpose of this study is to describe the continuing education program and summarize participants' learning outcomes and perceptions of the program. This year-long virtual program is a model that could be adopted and adapted by other institutions as well as across institutions given its virtual format.

2. Methods

2.1. Program goals and theoretical framework

The yearlong virtual continuing education program was directed by faculty with experience in health professions education consisting of 2 physicians and 1 physician assistant. Administrative support was provided by 2 program coordinators and program evaluation was managed by the director of assessment and evaluation holding an education doctorate. The primary goals of the yearlong program were for participants to.

1. Describe the need and rationale for integration of the principles of interprofessional collaborative practice into all health professions education.
2. Increase knowledge and skills required to advance interprofessional education and education program planning.
3. Apply knowledge of educational principles, program design, instruction, and evaluation through development of an interprofessional education experience.

The theoretical framework for this program design was based on experiential learning and social learning paradigms. Experiential learning paradigms emphasize individualized learning based on participants' actual lived experiences and contexts.¹⁴ Social learning paradigms focus on learning situated in the context and community of the individual learner.¹⁵ These two paradigms informed instructional methods which involved individual and group learning with reflection on current practice plus application of new knowledge and skills.

2.2. Target audience

The target audience for the program included health professions educators across the range of professions and experience levels interested in developing skills to design and implement IEEs. Roles were inclusive of but not limited to clinical instructors, clinical preceptors, course directors, clerkship directors, program directors, curriculum deans, or interprofessional clinical team leads. Eligible participants applied as individuals or teams then were selected to participate based on specific criteria by the program directors and a committee of experienced educators from multiple health professions. Selection criteria included years of experience as a health professions educator and the quality interprofessional educational experience proposal, including how it would advance interprofessional education and the level of collaboration with other health professionals in its design. Participants incurred no direct cost for participation as the program was funded by the institution's academy for health professions educators and center for interprofessional education and care. Participants were expected to attend all sessions and were eligible for a stipend to cover qualifying expenses related to the IEEs. Participants were eligible to claim continuing education credits for sessions attended to a maximum of 40

category 1 h. Credits were approved for awarding through multiple organizations to ensure participants from various professions could claim credits, including American Medical Association (AMA), International Accreditors for Continuing Education and Training (IACET), and Joint Accreditation (JA). Applicants applied individually and also as part of a team, which would be planning collaboration on a specific interprofessional education experience. The program aimed to select around 20 applicants based on the availability of project mentors, small group activities, planned participant presentations, and prior success in other programs with similar size cohorts. The program selected 19 of 36 individual applicants, of which 16 were on 1 of 9 teams.

2.3. Schedule and instructional activities

The schedule for the yearlong program consisted of once monthly 2-h virtual meetings for interactive presentations and discussion plus an additional 2–4 h per month for individual or team-based IEE planning and supplemental learning opportunities (e.g., small group discussion, mentoring). Virtual meetings were conducted on the institution's secure videoconferencing platform and program-related correspondence and content were provided by e-mail. Monthly virtual meetings consisted of preparatory activities (e.g., readings), interactive presentations, and small group discussion. Preparatory activities consisted of assigned readings from the program primary textbook or articles from the health professions education literature.¹⁶

Interactive presentations included those provided by experts in the topical areas as well as those delivered by program participants on their self-selected *deep dive* topic and IEE summaries. Participants' presentations were completed in the last 3 months of the program. Inter-session activities consisted of individual assignments and group activities. Individual assignments were primarily related to research on the *deep dive* topic of interest, preparation for monthly sessions, and individual reflection. Group activities included engagement with other program participants and colleagues in their clinical practice settings to advance their IEE planning. Participants also engaged in at least one mentoring session with the program directors to discuss their ongoing professional development and for guidance on their IEE planning.

2.4. Instructional content

Instructional content for the program was derived from the primary textbook, expert presenters, and program directors, as well as topics selected by participants for *deep dive* presentations and summaries of planned IEEs. Expert presenters included doctoral-trained educators (e.g., PhD, EdD), medical librarians, nursing professionals, occupational therapists, physician assistants (PAs), physical therapists, and physicians. Instructional content by expert presenters included topics related to professional development to support individual growth and curriculum development to support IEE planning. See Table 1 for list of synchronous content topics by category. Expert presenters were supported by the program directors through individual meetings which included an orientation to the program, review of a standardized framework for presentations, and discussion of best practices in delivering

Table 1
Virtual synchronous session content by type.

Individual Professional Development	Interprofessional Education Experience Design
Educator Identity & Development	Principles of Adult Learning
Interprofessional Leading; Educational Theory	Problem Identification and Needs Analysis
Educational Literature Review	Learning Goals and Objectives
Presenting and Providing Peer Feedback	Instructional Methods
Discovery and Dissemination	Assessment and Evaluation
Diversity, Equity, and Inclusion	Management and Implementation
Self-Care and Resilience	Management and Implementation

presentations. Instructional content was also presented by program participants on topics of interest for which they would conduct an in-depth analysis. The purpose of these *deep dive* presentations was to increase participants' knowledge about a topic of interest, share new information with fellow participants, practice presentation skills, and receive peer feedback on their presentation skills.

2.5. Design of interprofessional education experiences

One major goal was for participants to apply leadership principles and skills in collaboration with other professions to design an IEE for health professions learners or providers. Leadership was a singular topic for one synchronous session and a theme addressed across multiple expert presentations. Potential IEEs could include short courses, workshops, seminars, or project-based initiatives. IEEs were expected to have multiple professions involved in the planning process and have an interprofessional audience. In alignment with institutional values, participants were also encouraged to account for diversity, equity, and inclusion in all stages of design and delivery, which was also a focus of an expert presentation. As a product of this design process, participants developed a detailed implementation and evaluation plan which they presented at the end of the program. Participants were encouraged to implement the IEE either during or after the program, though this was not a requirement for participation. Design of the IEEs was supported through topic presentations based on the program textbook, mentorship from the program directors, and collaboration with fellow participants and other colleagues from their clinical practice setting. Participants were also provided with a planning document with guiding questions to assist in this process, summarized in Table 2. Implementation of IEEs was incentivized and supported through eligible funds of \$750 per participant for qualifying expenses (e.g., instructional materials, food, evaluation consultation, statistical support).

2.6. Program evaluation

Primary program outcome measures for evaluation consisted of the number of participants who (1) completed the program, (2) presented a *deep dive* presentation, and (3) designed an IEE. Participant achievement of these outcomes were measured by attendance throughout the program (defined as attendance at least 10 of 12 sessions attended) and observed presentations of *deep dive* topic and summary of the designed IEE. Survey data were collected using the Qualtrics platform, responses were anonymous, and each participant created a unique identification code to allow for comparison of survey data. At the beginning and end of the program, participants' perceptions of change in knowledge, skills, and attitudes toward interprofessional program planning were also assessed with a survey.

Survey items were adapted from an international study of in-person faculty development workshops designed to improve knowledge, skills, attitudes, and practice in interprofessional continuing education.⁶ This survey was selected based on similarity in goals between the in-person

workshops described in that study and our new virtual-based program. For future program improvement, the survey also assessed participants' satisfaction with the program, instructional content, instructional methods, and content topics. Approximately 1 year after program completion, participants were invited to complete a brief survey to provide an update on implementation of the IEE they designed during the program.

Survey data was examined using quantitative analysis of numeric data plus qualitative analysis of written responses. Quantitative methods were completed using Excel 2021 (Microsoft, Redmond WA) and included descriptive statistics and inferential analysis. Quantitative results are reported in aggregate format with frequencies and percentages. Inferential analysis using paired t-tests was conducted to identify statistically significant changes in participants' perceptions of knowledge and skills before and after the program. The significance level was set at $P < .05$. Qualitative analysis of short written responses in surveys are also summarized. More robust qualitative methods were not used given the limited number and length of responses. The Institutional Review Board determined that this study was exempt from review as education focused non-research.

3. Results

Nineteen participants were selected to participate in the program from the following professions: registered and advanced practice nurses (6), chaplain services (1), occupational therapy (1), physical therapy (1), physician/medicine (9), and social work (1). The 3 surveys had the following response rates: pre-program survey 100 % (19/19), post-program survey 53 % (11/19), 1-year follow-up survey 74 % (14/19). All 19 participants (100 %) successfully completed the program based on attendance of at least 10 of 12 synchronous sessions. All participants (100 %) also met program goals to present on a medical education topic of interest and their designed IEE. All respondents (100 %) to the end of program survey reported meeting their personal and professional goals for the program. Respondents' perceptions of knowledge, skills, and attitudes in the pre-program versus post-program surveys are reported in Table 3. Respondents' self-efficacy in knowledge and skills/competencies in interprofessional education increased significantly during the program. There were not statistically significant changes in perceived attitudes.

All participants achieved the program requirement to develop an IEE as individuals or as part of a group. See Table 4 for title of IEEs and respective target audiences. While not a program requirement, nearly half of respondents (45 %) to the end of program surveyed reported implementing their IEE during the program. In the 1-year follow-up survey, more than half of the respondents (57 %) reported implementing the IEE during the program or in the year after completion of the program. A single respondent planned to implement the IEE 1-year after program completion. Those who had not implemented the IEE cited several barriers, including time commitment, other responsibilities, expenses, limitations due to other curricula, and difficulty collaborating

Table 2
Interprofessional education experience plan with guiding questions.

Educational Problem or Need: What educational problem or need have you chosen as the focus of your program planning? Why is this program needed in your institution? What curricula already exist in at your institution or other institutions? Do you need to conduct a general or targeted needs assessment?
Target Learners: Who are your target learners and professions? Please describe their existing proficiencies, current performance, attitudes, perceived deficiencies and needs.
Goal(s) of the Interprofessional Education Activity: What is the overall purpose of the interprofessional education activity? What is the intended impact? Consider impact on participants and who they serve, including learners and patients.
Learning Objectives: What would you like the learners to achieve through the participation in your program? What will they know/be able to do at the end of your program?
Instructional Content: List content areas to be taught as part of the program in order to assist learners to achieve the outcomes
Educational/Instructional Strategies: List events/teaching methods you will use to teach the content
Evaluation Methods: List evaluation methods you will use to assess if the learning outcomes were achieved
Barriers: What barriers/limitations you might encounter and how you might overcome them?
Stakeholder Involvement: Who are the stakeholders (e.g. patients, clinicians, administrators, educators) related to this interprofessional education activity? How should you involve them in this interprofessional education activity?
Budget: What are the expected costs associated with this interprofessional education activity? Are the costs justified? What are potential sources of funding?

Table 3

Change Pre-Program and Post-Program Perceptions of Participants using Paired, Two-Tailed T-Test (n = 10).

Domain	Item	Pre-Program Mean	Post-Program Mean	p-value
Knowledge	I know the definition of interprofessional education	3.5	5	<0.001
	I am familiar with interprofessional education competencies	2.7	4.8	<0.001
	I can identify when an interprofessional education activity is appropriate	2.7	4.6	<0.001
	I can list possible barriers to implementing interprofessional education in my practice setting	2.5	4.8	<0.001
	I am able to design interprofessional education activities	2.2	4.4	<0.001
Skills	I am able to facilitate a group of interprofessional learners	2.4	4.6	<0.001
	I am able to work collaboratively as a member of an interprofessional planning team	3.8	4.8	0.01
	I am able to incorporate the patient's perspective in planning interprofessional education	3.3	4.6	0.01
	I believe interprofessional education is important for improving team performance	4.6	5	0.10
	I believe interprofessional education can have a positive impact on patient outcomes	4.7	4.9	0.44
Attitudes	I believe interprofessional education is too difficult to implement	1.5	1.9	0.51
	I would welcome the opportunity to be a member of an interprofessional planning team	4.7	4.5	0.56

Table 4

Interprofessional education experience title and target audience.

Title	Target Audience
Cross-training Faculty for Interprofessional Education of Physical and Occupation Therapy Students	Educators
Telehealth Education	Learners
Interprofessional Approach to Standardizing Family Care During Code Blue Events	Providers, Staff
Code Blue Simulation Education for the Interprofessional Team	Providers, Staff
Education for Hospice Team	Providers, Staff
Interprofessional Passport for Outpatient Internal Medicine	Providers, Staff
Intramuscular Injection for Vaccine Administration COVID-19 Vaccine Clinic	Providers, Staff
Using Simulation for Postpartum Hemorrhage Education	Providers, Staff
Extracorporeal Membrane Oxygenation (ECMO) Training	Providers, Staff, Learners
Sepsis Protocol Education Training in Inpatient Settings	Providers, Staff, Learners

with more than 1 educational program. Participants rated program activities in the post-program survey which are summarized in Table 5. Recommendations for program improvement were identified based on analysis of free text responses with the following themes: less synchronous sessions, more synchronous time to focus on IEE development,

Table 5

Preferred program activities in rank Order (n: 11).

Program Activity	Frequency	Percentage
Synchronous Presentations on Individual Professional Development	11	100 %
Mentoring on Individual Professional Development	10	91 %
Synchronous Presentations on Individual Professional Development	10	91 %
Receiving Feedback on Presentation About Interprofessional Education Activity	10	91 %
Mentoring for Interprofessional Education Activity Development	9	82 %
Receiving Feedback on Presentation About Topic of Interest ("Deep Dive")	8	73 %
Session Readings and Resources	8	73 %
Small Group Discussions	5	55 %

allow for funding upon request.

4. Discussion

4.1. Program success in a virtual format

We successfully implemented a year-long virtual continuing education program with health professions educators from multiple professions focused on individual development and design of IEEs. Such programs have traditionally occurred in-person with a limited number of virtual offerings^{6,8}. We applied elements of experiential and social learning theory to create an engaging and efficient virtual learning experience for busy educators that may have not been accessible through an in-person program and in the context of the beginning of the COVID-19 pandemic. Every participant achieved the primary program goals as evidenced by high session attendance, participation in session activities, an individual presentation of *deep dive* topic, and an individual or group presentation about the designed IEE. Post-program survey results indicated statistically significant increases in knowledge and skills related to interprofessional education. Development of knowledge, skills, and attitudes have been identified as measures of success in continuing education programs for health professions educators.¹⁷ The lack of change in attitudes may be explained by highly positive pre-program attitudes which would be expected from educators interested in participating in such a program. The results from our program demonstrate the effectiveness of using virtual platforms for continuing education programs. This program has not been repeated, though we plan to implement it in an all-virtual or hybrid virtual-predominant format in the future. Moving to a hybrid model would help with in-person team building and possibly project completion, though it requires greater logistical, administrative, and cost support.

4.2. Advancing interprofessional education

This program was successful in recruiting health professions educators from multiple professions to design unique IEEs for health professions learners and providers. Prior reviews have indicated the need for faculty development programs to reach a more diverse audience, including clinician-educators from different professions working in both academic and clinical settings with a focus on real projects.¹⁸ Participants in this program represented 7 different professions and nearly half enrolled as a member of an IP team. All participants, whether individuals or members of a team, designed and presented an IEE by the end of the program. More than half (57 %) of respondents to the 1-year follow-up survey reported successful implementation one year after program completion. Additionally, one team even published results of their IEE in the health professions education literature.¹⁹ Program participants' achievement of these primary program outcomes are a realization of our institution's mission to advance interprofessional

education and collaborative practice.

Program participants' design of the IEE required application of newly acquired skills in curriculum development. Key elements of successful professional development programs include training in education program design and leadership so were included in the program instructional content.^{4,5} In addition to readings and expert presentations, participants refined IEE plans through regular virtual mentoring meetings with course directors and formal presentations to peers to solicit feedback. Participants indicated in the final survey that the opportunity to work on the IEE and to access the expertise of both classmates and directors was critical to their success. Participants also described barriers to implementing IEEs, including lack of time, identifying collaborators, and difficulty accessing space in the curriculum. Interestingly, several participants reported that they did not spend funds allotted for use on their IEEs and very few identified a lack of funding as their reason for not implementing their programs. Instead, most respondents identified a need for more time with mentors getting direction on IEE design and implementation. To better support IEE completion in the future, we plan to incorporate timelines for task completion, more accountability from participants' collaborators, more mentoring time with program directors, and financial support for specific IEE expenses.

4.3. Individual and community development

The program provided instruction on topics intended to promote personal growth and leadership and build rapport and trust among participants. These included sessions focused on leadership, interprofessional teamwork, self-care as well as diversity, equity, and inclusion. These topics were recognized needs within our institution and also identified in a study by another institution.²⁰ Development of leadership skills is recognized as a key component of medical faculty development programs.^{18,21} In this program, applicants were able to apply these skills through IEE design and, for most participants, implementation. We also facilitated engagement among/between participants through mixing professions for small group work and providing an opportunity for every participant to do an individual *deep dive* presentation and receive feedback from their peers. As a result, participants were able to develop strong relationships and mutual trust with one another over the 12-month period and within the virtual format. Participants expressed they were able to establish communities of practice which is often missing for educators working in different professions, disciplines, and physical locations in a large academic health center.^{2,3,22} Future iterations of this program will continue to include time dedicated to both individual and development of communities of practice. For individuals, we plan to continue provision of guidance from program directors and add an expectation for participants to identify mentor from their practice setting to support the design of their IEE. For community development, we plan to continue to encourage teams to apply to participate and be more deliberate in assigning participants to the same small discussion groups throughout the course of the program.

4.4. Limitations

The evaluation of the implementation and outcomes of this program has several limitations. This report has limited generalizability given the relatively small number of program participants, few demographic descriptors of program participants, and location at a single institution. Furthermore, there were no prior iterations of this program which to compare the outcomes of this inaugural program. Several of the survey items adapted for this program covered key domains for evaluation, but has not been validated as a tool to assess change in knowledge, skills, and attitudes among clinician-educators.⁶ While survey feedback identified participants' value of collaborative group activities, socialization and collaboration was not formally evaluated. The authors did not compare survey results from this program to the other study because survey was not validated and it was used for a different continuing

education program in terms of format and participants. The authors had initially planned to include qualitative methodology in program evaluation, though they could not recruit enough participants to participate in a post-program focus group. This study was not designed to assess individual factors associated with program completion or implementation of interprofessional education experience, though the authors did aim to understand barriers to project implementation through surveys. Finally, social desirability bias may have influenced survey responses despite efforts to mitigate this effect by making surveys anonymous.

5. Conclusion

This study provides evidence of the effectiveness of a virtual platform for longitudinal continuing education programs aiming to advance individual professional development and the design of IEEs. The online platform supported the individuals and teams in their planning for IEEs while promoting individual development and creating a community of learning. Most participants implemented IEEs leading to the desired real-world impact and helping achieve the institution's strategic priority to advance interprofessional education and care. We plan to offer this program in the upcoming year with improvements based on feedback from participants and guidance from emerging literature on virtual-based continuing education programming. Future research is needed to identify optimal instructional methods for longitudinal virtual programs, understand the limits of virtual programs compared to in-person programs, and examine the long-term impact of participation on interprofessional education and care. The virtual interprofessional continuing education program described in this study could be adapted within and across institutions in support of health professions educators leading interprofessional education and collaborative practice initiatives.

CRedit authorship contribution statement

Nicholas M. Hudak: Writing – review & editing, Writing – original draft, Visualization, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Mitchell Tod Heflin:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Conceptualization. **Diana McNeill:** Writing – review & editing, Supervision, Resources, Project administration, Methodology.

Declaration of competing interest

Based on reviewer feedback, I am writing to submit a revised version of the original manuscript, entitled *Increasing Interprofessional Education Experiences and Collaboration: Outcomes of a Virtually-Based Continuing Education Program*, for further consideration as an original research article.

I certify that all authors have been included on the original manuscript submission and meet all three of the following conditions for authorship credit: (1) substantial contributions to conception and design, acquisition of data, or analysis and interpretation of data; (2) drafting the article or revising it critically for important intellectual content; and (3) review of final version and approval of the final version to be published.

I certify that the enclosed manuscript represents original work and that neither this manuscript nor a manuscript with substantially similar content under our authorship has been published or is currently being considered for publication by any other publication. This manuscript will not be submitted to another publication while it is under consideration by the Journal of Interprofessional Education and Care.

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I further understand that we are solely responsible for the content of the published article, including changes made during editing; we will receive a copy of the edited manuscript to review and approve prior to publication.

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